

Ameet Rao's Calculus III Notes

Cylindrical and Spherical Coordinates and Converting Between Systems

sph to cyl.

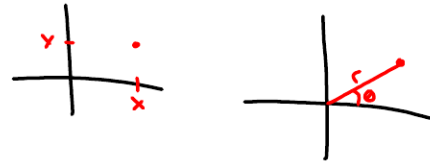
$$r^2 = \rho^2 \sin^2 \phi \quad \theta = \phi \quad z = \rho \cos \phi$$

cyl. to sph.

$$\rho = \sqrt{r^2 + z^2} \quad \theta = \phi \quad \phi = \arccos\left(\frac{z}{\sqrt{r^2 + z^2}}\right)$$

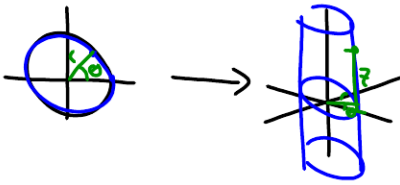
Switching between the two systems: θ never changes. Going out, $r = \rho \sin \phi$ and $z = \rho \cos \phi$. Coming back, $\rho = \sqrt{r^2 + z^2}$ and $\phi = \arccos(z/\sqrt{r^2 + z^2})$.

$$(x, y) \rightarrow (r, \theta)$$



Polar coordinates describe a point by its distance r from the origin and its angle θ from the x -axis, instead of by (x, y) .

Cylindrical coordinates
 $(x, y, z) \rightarrow (r, \theta, z)$



Cylindrical coordinates (r, θ, z) are polar coordinates plus the same height z . Fixing r traces a cylinder around the z -axis.

$$(x, y, z) \rightarrow (r, \theta, z)$$

$$\begin{aligned} x &= r \cos \theta \\ y &= r \sin \theta \\ z &= z \end{aligned}$$

Cylindrical to rectangular: $x = r \cos \theta$, $y = r \sin \theta$, $z = z$. The height doesn't change.

$$(r, \theta, z) \rightarrow (x, y, z)$$

$$\begin{aligned} r^2 &= x^2 + y^2 \\ \tan \theta &= \frac{y}{x} \\ z &= z \end{aligned}$$

Rectangular to cylindrical: $r^2 = x^2 + y^2$, $\tan \theta = y/x$, $z = z$. Check which quadrant the point is in before choosing θ .

$$\textcircled{1} (4, \frac{5\pi}{6}, 3) = (-2\sqrt{3}, 2, 3)$$

$$(r, \theta, z) \rightarrow (x, y, z)$$

$$\textcircled{2} (1, \sqrt{3}, 2) = (2, \frac{\pi}{3}, 2) \text{ or } (-2, \frac{4\pi}{3}, 2)$$

$$(x, y, z) \rightarrow (r, \theta, z)$$

$\textcircled{1}$ $(4, 5\pi/6, 3)$ becomes $(-2\sqrt{3}, 2, 3)$. $\textcircled{2}$ $(1, \sqrt{3}, 2)$ has $r = 2$ and $\tan \theta = \sqrt{3}$, so $(2, \pi/3, 2)$. Equivalent: $(-2, 4\pi/3, 2)$, using a negative r .

Ameet Rao's Calculus III Notes

Cylindrical and Spherical Coordinates and Converting Between Systems

$$y^2 = x \quad \text{Find cyl. eq.}$$

$$(r \sin \theta)^2 = r \cos \theta$$

$$\underline{r^2 \sin^2 \theta = r \cos \theta}$$

$$r \sin^2 \theta = \cos \theta$$

$$r = \frac{\cos \theta}{\sin^2 \theta} = \cot \theta \csc \theta \quad r \neq 0$$

Convert $y^2 = x$: substitute $y = r \sin \theta$ and $x = r \cos \theta$, so $r^2 \sin^2 \theta = r \cos \theta$. Divide by r : $r = \cos \theta / \sin^2 \theta = \cot \theta \csc \theta$ ($r = 0$ is also on the surface).

$$r^2 \cos^2 \theta + z^2 + 1 = 0$$

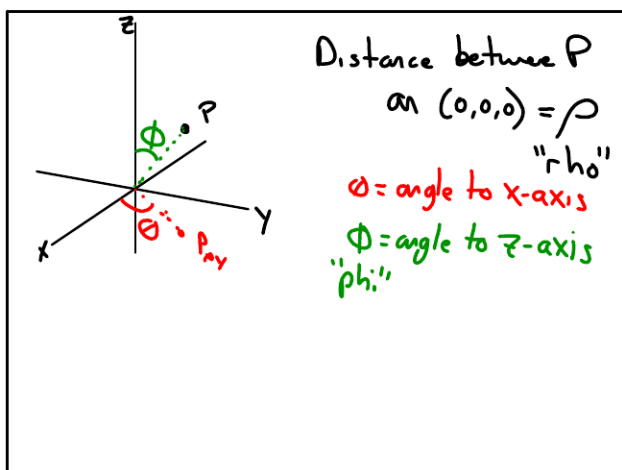
$$r^2 (\cos^2 \theta - \sin^2 \theta) + z^2 + 1 = 0$$

$$\underline{r^2 \cos^2 \theta - r^2 \sin^2 \theta + z^2 + 1 = 0}$$

$$-x^2 + y^2 - z^2 = 1$$

$$y^2 - x^2 - z^2 = 1$$

Going back: $r^2 \cos^2 \theta = x^2$ and $r^2 \sin^2 \theta = y^2$. So $r^2 (\cos^2 \theta - \sin^2 \theta) + z^2 + 1 = 0$ becomes $y^2 - x^2 - z^2 = 1$, a hyperboloid of two sheets.



Spherical coordinates (ρ, θ, ϕ) : ρ is the distance from the origin, θ is the angle from the x -axis (same as before), and ϕ is the angle down from the z -axis.

$$(x, y, z) \rightarrow (\rho, \theta, \phi)$$

Spherical coordinates locate a point by ρ (distance out) plus two angles, θ and ϕ . The next two boxes give the conversion formulas.

spherical to rectangular

$$x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

$$z = \rho \cos \phi$$

Spherical to rectangular: $\rho \sin \phi$ is the distance out from the z -axis, which $\cos \theta$ and $\sin \theta$ split into x and y . The height is $z = \rho \cos \phi$.

rectangular to spherical

$$\rho^2 = x^2 + y^2 + z^2$$

$$\tan \theta = \frac{y}{x} \quad \theta = \arctan\left(\frac{y}{x}\right)$$

$$\phi = \arccos\left(\frac{z}{\sqrt{x^2 + y^2 + z^2}}\right)$$

$$\cos \phi = \frac{z}{\rho}$$

Rectangular to spherical: $\rho^2 = x^2 + y^2 + z^2$ (3D distance), $\tan \theta = y/x$, and $\phi = \arccos(z/\rho)$, since $\cos \phi = z/\rho$.